Fiscal Year:	FY 2013	**Task Last Updated:**	FY 03/11/2013
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:	https://		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2012	**End Date:**	05/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 5/31/2017 per NSSC information (Ed., 6/17/13) NOTE: Change in grant number and to be extended, per D. Chao/GRC (Ed. 3/6/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AB01G ; NNX12AK14G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the manned mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for ISS experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
Task Progress:	Vehicles for future manned space missions will demand unprecedented increases in power requirements and heat dissipation. Achieving these goals while maintaining acceptable size and weight limits will require replacing present single-phase thermal management components with far more efficient two-phase counterparts. This study discusses the development of an experimental facility for the study of annular condensation of FC-72 in microgravity, which was tested in parabolic flight as a prelude to the development of NASA's Flow Boiling and Condensation Experiment (FBCE) for the International Space Station (ISS). The flow behavior of the condensate film is shown to be sensitive mostly to the mass velocity of FC-72, with low mass velocities yielding laminar flow with a smooth interface, and high mass velocities turbulent flow with appreciable interfacial waviness. A select number of tests repeated in microgravity, Lunar gravity, and Martian gravity prove that the influence of gravity is very pronounced at low mass velocities, manifest by circumferential uniformity for microgravity versus appreciable thickening along one side of the condensation tube for Lunar and Martian conditions. However, the thickening is nonexistent for Lunar and Martian conditions at high mass velocities due to increased vapor shear on the film interface, proving high mass velocity is an effective means to negating the influence of gravity in space missions. For microgravity, the condensation heat transfer coefficient is highest near the inlet, where the film is both thin and laminar, and decreases along the condensation length, but increases again downstream for high mass velocities due to turbulence and increased waviness. A model is proposed to predict the condensation heat transfer which accounts for dampening of turbulent fluctuations near the film interface. The model shows good agreement with the heat transfer coefficient data in both trend and magnitude.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Abstracts for Journals and Proceedings	Mudawar I, Hasan MM. "Flow Boiling Critical Heat Flux in Reduced Gravity." 28th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), New Orleans, LA, November 28-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), New Orleans, LA, November 28-December 2, 2012. Program and Abstracts, p. 53. https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Dec-2012
Abstracts for Journals and Proceedings	Mudawar I, Hasan MM. "Development of Flow Boiling and Condensation Experiment (FBCE) for the International Space Station." 28th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), New Orleans, LA, November 28-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), New Orleans, LA, November 28-December 2, 2012. Program and Abstracts, p. 53. https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Dec-2012
Articles in Peer-reviewed Journals	Kim S-M, Mudawar I. "Universal approach to predicting two-phase frictional pressure drop for mini/micro-channel saturated flow boiling." International Journal of Heat and Mass Transfer. 2013 Mar;58(1-2):718-34. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2012.11.045 , Mar-2013
Articles in Peer-reviewed Journals	Lee H, Mudawar I, Hasan MM. "Experimental and theoretical investigation of annular flow condensation in microgravity." International Journal of Heat and Mass Transfer. 2013 Jun;61:293–309. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2013.02.010 , Jun-2013